

SMART DEVICE ROBOT DIARY SYSTEM

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Abstract:

This paper presents the development of a mobile diary application in conjunction with a robotics-based smart device. This system was designed to address the challenges faced by elderly individuals, such as loneliness and social isolation, which can result from low self-efficacy. The app comprises a graphical user interface, an experience extraction model, a self-growing memory agent, a chat agent, and a feedback interactive. The latter processes diary entries from elderly users, generating empathic responses based on successful experiences that match the individual's capabilities. This is done while learning and adapting to their behavioral patterns and emotional states. By recognizing the user's emotions and history of human-like interactions, and by integrating user feedback, the system aims to motivate the behavior of elderly people, thereby increasing their self-efficacy and improving their overall well-being.

Keywords:

Applications; Intelligent Robot; Human-Centric Systems; Agent Based Technologies

1. Introduction

The global population is aging, which presents a set of distinctive challenges in countries across the globe. One of the most significant issues is the prevalence of loneliness and social isolation among the elderly. These conditions not only reduce the quality of life of the elderly but are also associated with various mental and physical health issues. There has been a notable increase in interest surrounding the potential of artificial intelligence (AI) to address these issues. Moreover, in light of the growing scarcity of labor, AI-driven innovations are assuming a more prominent role within the global elderly care sector. An increasing number of regions are turning to AI technology as a means of addressing the shortage of personnel available to provide care for the elderly.

The term "dying alone" has been coined to describe the growing phenomenon of elderly individuals dying alone and

unnoticed for extended periods of time [1,2]. In response to this trend, various efforts have been undertaken by governments and other stakeholders to leverage AI technologies to address this issue. These efforts have included the implementation of automated voice interaction systems and sensory robots, which aim to provide timely feedback and companionship, as well as enhance daily interactions for the elderly. A noteworthy example is the humanoid robot "Pepper," developed by SoftBank Robotics. Pepper is capable of understanding and responding to human emotions, rendering it an appropriate companion for the elderly [3]. Although studies have indicated that this AI technology can enhance the mental wellbeing of the elderly, further research in this field is ongoing.

A number of countries, including the United Kingdom and Sweden, have implemented artificial intelligence technologies in care facilities. These include voice-activated smart speakers that facilitate instant communication between elderly residents and staff members through responsive voice interaction. Such technologies foster a sense of control and motivation in elderly individuals. It is crucial to underscore that while AI has demonstrated efficacy in addressing loneliness among the elderly, it should be utilized as an adjunct to, rather than a replacement for, human interaction.

In the context of AI and loneliness, the concept of self-efficacy [4], which refers to an individual's belief in their ability to accomplish specific tasks, becomes highly relevant. Derived from Bandura's social cognitive theory, self-efficacy can impact a person's willingness to engage with AI technologies and, as a result, influence their capacity to address feelings of loneliness [5].

Despite the considerable potential of AI in addressing the challenges faced by the elderly, the existing Natural Language Processing (NLP) technologies still encounter significant

obstacles. These include the accuracy of semantic understanding, the naturalness of dialog generation, and the adaptability to cultural and geographical differences. However, the advent of large language models (LLMs) offers a novel solution to these challenges [6]. LLMs are capable of a more profound comprehension of intricate linguistic structures and human emotions, thereby facilitating the generation of conversations that are more natural and contextualized.

Furthermore, it is imperative to underscore the significance of tailoring counsel to the specific capabilities of the elderly. Such an approach not only assists the elderly in maintaining necessary social interactions but also encourages them to engage in new activities within their capabilities, thereby facilitating the gradual expansion of their social networks and life experiences while avoiding suggestions that exceed their actual capabilities. Incompatible suggestions may not only result in task failure but may also have a detrimental impact on the elderly's sense of self-efficacy. It is therefore important to provide tasks that are neither too easy nor too difficult, and to set achievable goals.

The use of AI to enhance the well-being of the elderly, particularly in terms of reducing loneliness and fostering self-efficacy, has been extensively documented in academic literature. However, the system proposed in this paper represents a novel development distinct from existing approaches in the field.

The objective of this paper is to present a system that employs the use of a diary application with the intention of enhancing the sense of self-efficacy among the elderly population. The act of maintaining a diary enables users to gain insight into their own behavior patterns, which can subsequently inform the formation of habits. By recording positive events and their associated emotions, individuals can identify patterns in their lives and replicate those that are beneficial. A multitude of diary applications for mobile devices have been developed; however, they are typically solitary tasks. Therefore, our objective is to utilize a robot to transform this activity into a collaborative endeavor, ensuring its continued usability.

2. Related works

2.1. Diary app research

A variety of digital diary applications have been developed with the objective of assisting users in the management of their health. These applications are

designed to facilitate the recording of various health-related events and behaviors, including headaches [7], dietary habits [8], cognitive behavioral therapy (CBT) [9], and diabetes management [10]. The benefits of journaling for mental and emotional well-being have been well-documented, and counselors, social workers, and therapists frequently recommend journaling as a tool for their patients [11]. It was reported that participants were able to process emotional events such as trauma and stress by engaging in a brief writing exercise, comprising 15 to 20 minutes of writing three to five times. Mood tracking apps have demonstrated that users utilize mood tracking to gain insight into their mood patterns, enhance their mood, and self-managed mental illnesses [12].

2.2. Diary app with a communication robot

Diaries are also utilized as applications for commercial robots sold in Japan, functioning as a medium for interaction between robots and humans. Sharp's RoBoHoN [13] has the capacity to record the events of the day in a manner analogous to a diary when engaged in conversation with the robot. Despite the inability of GROOVE X, Inc.'s LOVOT [14] to speak, its mobile application offers a diary feature that displays a timeline of human interactions and basic actions.

In the European ALIZ-E project, the NAO robot has been demonstrated to motivate children with diabetes to maintain a daily record. Prior to utilizing the robot, none of the children maintained a daily diary [20]. However, following its introduction, they engaged in diary-keeping at a markedly higher frequency. Furthermore, a method has been proposed whereby the robot itself maintains a diary. Additionally, experimental studies have demonstrated that verbalizing a robot's memories in the form of a diary can positively influence readers' perceptions. Kochigami et al. demonstrated that robots can facilitate the formation of friendships with children by presenting them with an open diary [15]. Furthermore, it has been demonstrated that the act of sharing a diary with children can facilitate communication between them and adults.

Ichikura et al. put forth an automatic diary generation system that leverages data from past joint experiences, with the objective of enhancing the appeal of a robot through the fostering of shared experiences between humans and robots [16]. A robot and a human engaged in a series of interactions, which were recorded in a diary that documented the history of these interactions and the dialogue that occurred. The proposed diary was rated highly in the categories of "comfort" and "performance" in the evaluation of the robot's impression. In a survey of diaries that would elicit a more

favorable impression, diaries that included information about joint experiences were rated higher than diaries that did not include such information. This is due to the fact that diaries comprising information regarding joint experiences indicate a greater degree of collaboration between the robot and the human, as well as a sense of intimacy from the robot.

3. Proposed system

The objective of this proposal is to present a robot diary system that employs the use of a robot in conjunction with a smart device. The utilization of a smart device enables the provision of robotics to a broader population.

3.1 iPhonoid

iPhonoid is a robot partner that utilizes the iPhone, a smart device. The iPhone is equipped with a variety of sensors, including accelerometers and GPS, which allow for the elimination of the need to install additional sensors on the robot. This results in a reduction in the cost of the robot itself. Furthermore, transforming the iPhone, which is a device with which users are intimately familiar, into a robot fosters a sense of familiarity with the iPhone. Furthermore, the iPhone's intrinsic functionality as a communication device enables seamless communication with external servers without the need for specialized configuration. This makes the iPhone an ideal edge device for integration into existing systems.

The iPhone installed in iPhonoid is removable. Therefore, if the user changes the area of activity, the iPhone itself can be relocated, and it is also possible to detach only the iPhone and attach it to another iPhonoid. Since it utilizes the same smart device, it has the advantage of being able to interact with the user using the same data even if it is attached to a different iPhone. For convenience, it is referred to as iPhonoid even if an iPad is used.

iPhonoid is a robot partner that utilizes the iPhone, a smart device. The iPhone is equipped with various sensors such as acceleration sensors and GPS, and by utilizing these sensors, it is not necessary to install independent sensors on the robot. This reduces the cost of the robot itself. Also, by making the iPhone, which users use every day, into a robot, it creates a sense of familiarity with the iPhone. Furthermore, since the iPhone itself is a communication device, it can communicate with external servers without special settings, and iPhone can be effectively used as an edge device.

The iPhone installed in iPhonoid is removable. Therefore, if the user changes the area of activity, the iPhone itself can be moved, and it is also possible to remove only the iPhone and

attach it to a different iPhonoid. Since it uses the same smart device, it has the advantage that it can interact with the user using the same data even if it is attached to a different iPhone. For convenience, it is called iPhonoid even if an iPad is used.



Fig. 1. iPhonoid

3.2 Diary app

The Diary app offers a user-friendly and straightforward interface for elderly individuals to input their diaries. This constitutes the inaugural stage in the documentation of the user's cognitive processes, affective states, and lived experiences in the form of a diary within the system. To this end, we have developed a Diary application that elderly users can implement on their tablet PCs or smartphones to input their emotions and experiences in diary form at their convenience. Furthermore, the application incorporates data visualization capabilities, enabling the examination of daily activities and mood fluctuations and displaying them in a graphical format for users.

Figure 2 provides an overview of the system usage, while Figure 3 depicts the actual image of the system in use. The graphical user interface (GUI) has been designed with an intuitive layout in order to facilitate ease of access and use for individuals. Subsequently, the original diary is verified and transferred to the emotion recognition model.

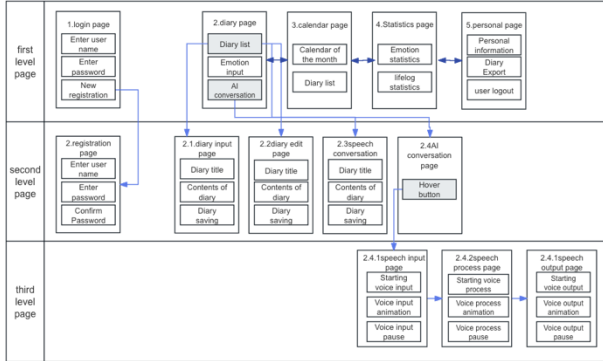


Fig. 2. How to Use the Diary Application



Fig. 3. Diary Application Interface for Actual Use

3.3 Experiment

The system is capable of recognizing emotions conveyed in the diary entries. However, since the diary lacks the functionality to express emotions, the robot employs facial expressions as a means of communication. The robot conveys emotional states and displays empathy through verbal and nonverbal communication. Figure 4 depicts a scene in which a diary is being written.



Fig. 4 Writing diary with keyboard and voice

In regard to the emotion recognition model utilized in the diary, the "cardiffnlp/twitter-roberta-base-emotion-multilabel-latest" model [17] is employed. This sophisticated model employs deep learning methodologies to analyze textual data and ascertain the emotional context. The model is capable of classifying emotions into 11 distinct categories: anger, anticipation, disgust, fear, joy, love, optimism, pessimism, sadness, surprise, and trust.

This method of emotion recognition entails the analysis of the proportion of each emotion present in each text, with a resulting score. An illustrative example of the emotion recognition result is provided below.

Input: "In the morning, I tried a new dish I learned on TV. It was a little difficult, but I did very well."

TABLE 1 EMOTION RECOGNITION RESULT

No.	Label	Score
1	Joy	0.9478232264518738
2	Optimism	0.9429337382316589
3	Anticipation	0.5059658288955688
4	Trust	0.3456786274909973
5	Love	0.07204209268093109
6	Surprise	0.0705651193857193
7	Fear	0.02815101481974125
8	Anger	0.027503671124577522
9	Disgust	0.015379147604107857
10	Pessimism	0.010825438424944878
11	Sadness	0.008868584409356117

Once the scores for each emotion have been obtained, the system selects the emotion with the highest score, which is then included in the diary text. Subsequently, the robot

executes a gesture that is consistent with the identified emotion (in this case, joy) and provides counsel (see Figure 5).



Fig. 5 Emotion expression of the robot

3.4 Discussion

The utilization of a robot to augment motivation for journaling is not a novel approach. Nevertheless, it is imperative to assess the influence of a system that incorporates a journaling application and a robot on motivation, in comparison to a system in which the robot and the journaling application are distinct entities.

Furthermore, the implementation of a virtual agent has the potential to facilitate communication with the user through multi-embodiment [18]. The field of multi-embodied agent research offers insights into the fostering of a positive human-agent relationship. This approach entails the embodiment of robots and virtual agents with both physical and virtual bodies, thereby enabling them to transition seamlessly between real and virtual environments to address user needs. Luria et al. identified four distinct modes of artifact embodiment in multi-embodied systems [19]. The re-embodiment style represents a single social presence that transitions from one body to another, and will be employed in the system under discussion.

Ichikura et al. documented their experiences of walking with a robot, but the presence of a virtual agent allows users to access locations that are inaccessible to robots. Moreover, the user's workload is diminished by enabling the virtual agent to document experiences, while the user records only their immediate sentiments.

4. Conclusions

We put forth a proposal for the development of a digital diary application integrated with a robotic system designed for use with smart devices. The objective of utilizing a diary application is to enhance self-efficacy and mitigate feelings of loneliness. However, given that individuals can engage in journaling with a robot in lieu of a standalone diary application, it is anticipated that feelings of loneliness will be diminished. The recognition of the emotion, the diary content, and the robot's expression of emotion were all successful.

In the future, we intend to develop a multi-embodied diary system to provide users with a more efficient and comprehensive support structure.

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